Supplementary Material

for: Lesion-grounded prototype meta-learning for few-shot thyroid ultrasound: segmentation, diagnosis, and faithful explanations under domain shift

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This document preserves, verbatim, the methodological detail moved out of the 16-page main text. Each block is labelled and cross-referenced to the title of the main-text subsection that cites it. Main-text equations and section numbers referenced here are those of the compressed manuscript.

S1 Supplementary data, annotation, and de-duplication detail

S1.1 Per-image recorded fields (referenced from main text, Section 'Data sources and annotation roles').

The data pipeline records the following fields for every image:

- dataset and domain identifier;
- patient, nodule, or examination identifier, where available (TN3K, TNUI-2021, and DDTI do not provide usable patient linkage in the released files - a recorded limitation);
- benign/malignant label and its source;
- mask, bounding box, or no localisation annotation;
- image dimensions, colour mode, and acquisition metadata, where available;
- split assignment and episode membership; and
- a cryptographic (SHA-256) image hash for duplicate checking.

S1.2 Annotation hierarchy (referenced from main text, Section 'Data sources and annotation roles').

The following annotation hierarchy is maintained:

1. pixel masks supervise or evaluate segmentation;
2. bounding boxes provide weaker localisation annotation and define the oracle region of interest where masks are absent (TN5000);
3. image- or case-level pathology labels supervise malignancy classification; and
4. TI-RADS or radiologist-derived labels are reported separately from pathology-confirmed outcomes.

S1.3 Label parsers: full precedence specification (referenced from main text, Section 'Data sources and annotation roles'; the TN5000 malignant-if-any rule and the TNUI polarity assumption-with-evidence are retained in the main text).

Label provenance and harmonisation. Classification labels enter the pipeline through three documented parsers:

TN5000 (source). Pascal-VOC XML with object class names `"0"` (benign) and `"1"` (malignant). The image label is *malignant-if-any* across annotated objects, and the oracle ROI box is the union of the boxes of the objects that decide the label. The official `ImageSets/Main` split files are used as released (the combined `trainval` file is not used).

TNUI-2021 (external). Per-split CSV files (`ID`, `Labels`; numeric 0/1) with identical schemas across splits. The polarity (1 = malignant) is not stated in public documentation. We adopt this polarity as an *assumption-with-evidence*: the alternative polarity would force in-domain-validated features to anti-classify on every TNUI cohort, and cohort sizes and prevalence are consistent with the published dataset description. The assumption is recorded in the limitations rather than silently absorbed.

DDTI (external). Per-case XML diagnoses are parsed by an element-aware parser with a fixed precedence: literal *benign*/*malignant* strings override numeric diagnosis codes; diagnosis codes map 1 → benign and 3 → malignant (all other codes unknown); TI-RADS annotations map 1–3 → benign and $4^{(*)}/5$ → malignant. Evidence is aggregated malignant-if-any, mirroring the TN5000 rule, and cases without a parseable diagnosis are excluded and counted. A vote profile (how many cases decided by each evidence class) is printed in every run for audit.

S1.4 Exact-duplicate policy mechanics (referenced from main text, Section 'Data splitting, de-duplication, and external-validation protocol').

Exact-duplicate policy. Every image is hashed (SHA-256). Within each dataset, exact duplicate files are removed once per dataset, keeping the occurrence in the highest-priority partition (TN3K: trainval > test; TNUI-2021: train > val > test). Dropped duplicates are archived in a separate manifest. Cross-dataset hashing runs against every pool that donates support cases (TN3K training pool) and between every pair of classification cohorts. Exclusions in later phases replicate the exclusion policy locked by earlier phases, with a recompute fallback. Each experiment notebook independently re-derives the clean manifests from the raw dataset files. When the audit manifests from the dataset-audit phase are mounted, the re-derived manifests are verified against them at hash level. For classification phases, the TN5000 manifest must additionally re-derive the exact locked split sizes before any model is trained or applied.

S2 Supplementary model and training configuration

S2.1 Stage-1 supervised reference: optimisation and inference (referenced from main text, 'Stage 1 optimisation').

S2.0.1 Stage 1 optimisation and inference details (supervised reference)

The Stage 1 reference is a ResNet34-encoder U-Net. We train it with AdamW (learning rate 10^{-3}, weight decay 10^{-4}), mixed precision, and ReduceLROnPlateau scheduling (factor 0.5, patience 4), for at most 60 epochs with early stopping (patience 12). An exponential moving average (EMA) of the weights (decay 0.999) is maintained; each epoch selects the better of raw and EMA weights by validation Dice. Training-time augmentation (albumentations) comprises horizontal flip, affine scaling/translation (0.85–1.15, ±5%) with rotation (±10°), brightness/contrast jitter (±0.25), Gaussian (speckle-like) noise, mild Gaussian blur, and elastic deformation. At inference, we select a sigmoid operating threshold on the internal validation split only (sweep 0.10–0.70, step 0.02) and freeze it before any test or external evaluation. Test-time augmentation (TTA;

identity, horizontal flip, $\pm 5^\circ$ rotation; probability-averaged) is reported alongside single-view inference. The tuned threshold value is reported in Results.
S2.2 Stage-2 episodic meta-training regimen (referenced from main text, 'Stage 2: Episodic decoder meta-training').

S2.0.2 Stage 2: Episodic decoder meta-training (single source domain)

The encoder is frozen. Each training episode samples nested supports (K_m re-drawn uniformly from $\{1, 5, 10, 20\}$) and eight queries from the source training pool. We apply light *joint* augmentation (consistent flip, $\pm 10^\circ$ rotation, and per-image gain across all episode images and masks), compute multi-scale prototypes with the frozen encoder, and minimise the query loss with respect to the decoder parameters only. Optimisation uses AdamW (10^{-3}, 300 episodes per epoch) with early stopping on mean query Dice over 100 validation episodes (patience 8). Joint episodic meta-training across multiple source domains is a designated extension; the primary protocol trains on TN3K and characterises transfer to TNUI-2021 and DDTI under the zero-shot and mask-shot conditions defined above.
S2.3 Linear-head training configuration (referenced from main text, 'Classification heads').

Linear heads (primary). Each regime's descriptor is standardised with the training-split mean and standard deviation. We train a logistic linear head (a single affine layer) with AdamW (learning rate 3×10^{-3}, weight decay 10^{-2}), binary cross-entropy with positive-class weighting ($n_{\text{neg}}/n_{\text{pos}}$ on the training split), mini-batches of 512, at most 300 epochs, and early stopping on validation AUROC (patience 25, improvement margin 10^{-4}, seed 42). The only quantity that varies across regimes is *what is pooled*; standardisation, optimiser, and selection protocol are identical, so R0–R1p contrasts isolate grounding and R1p–R1g contrasts isolate segmentation error.
S2.4 Whole-image ERM baseline configuration (referenced from main text, 'Classification heads').

Whole-image ERM baseline (the fair comparator). The comparator is whole-image empirical risk minimisation (ERM). An ImageNet-pretrained ResNet34 with a single-logit head is fine-tuned end-to-end on TN5000 images (BCE with positive-class weighting; AdamW, learning rate 10^{-4}, weight decay 10^{-4}; cosine annealing over 15 epochs; early stopping on validation AUROC, patience 4; light augmentation: horizontal flip, $\pm 7^\circ$ rotation, gain jitter 0.8–1.2; seed 42). This row must be strong — it is the conventional alternative a reviewer compares the grounded pipeline against, both in-domain and externally. An uncertainty-aware (shrunk-covariance) prototype scorer is designated as an extension and is not part of the reported heads.
S2.5 Calibration configuration (referenced from main text, 'Stage 4: Calibration, operating points, and model selection').

S2.0.3 Stage 4: Calibration, operating points, and model selection

The source validation split (TN5000 val for classification; TN3K internal val for segmentation) is used for: early stopping; the segmentation operating-threshold sweep (validation only, then frozen); TTA configuration; decoder checkpoint selection; and, for classification, temperature scaling and the decision threshold. We fit a single scalar temperature per model by grid search over log-spaced candidates $T \in [0.05, 20]$, minimising validation negative log-likelihood on the model's validation logits. The operating threshold per model is the validation point of maximal sensitivity subject to specificity ≥ 0.90 on *raw* validation probabilities. Both are frozen before any test or external evaluation. No target-domain annotation is used for any of these decisions.
S2.6 Feature-bank per-case quality fields (verbatim from the feature-bank governing convention; referenced from main text, 'Lesion–context descriptor regimes').

Cached per case alongside the five descriptors: empty-mask flag, predicted area fraction, mean foreground probability, a 96×96 mask thumbnail.

S3 Extended formulation

S3.1 Episode and encoder definitions (referenced from main text, 'Episode formulation' and 'Shared multi-scale encoder').

A segmentation episode pairs a support set with a query set:

$$\mathcal{T}_{\text{seg}}^{(d)} = \left(\mathcal{S}_{\text{seg}}^{(d)}, \mathcal{Q}_{\text{seg}}^{(d)}\right), \tag{S1}$$

The support set contains K_m image–mask pairs and the query set $Q_m = 8$ disjoint images:

$$\mathcal{S}_{\text{seg}}^{(d)} = \{(x_i, m_i)\}_{i=1}^{K_m}, \qquad \mathcal{Q}_{\text{seg}}^{(d)} = \{(x_j, m_j)\}_{j=1}^{Q_m}, \qquad m_i \in \{0,1\}^{H\times W}. \tag{S2}$$

The shared encoder maps an input image to multi-scale feature maps:

$$\left\{F^l(x)\right\}_{l=1}^{L} = E_\theta(x), \qquad F^l(x) \in \mathbb{R}^{C_l\times H_l\times W_l}. \tag{S3}$$

S3.2 Uncertainty-aware segmentation prototypes (pre-specified extension, not executed; equations referenced from main text, 'Support-conditioned segmentation prototypes').

S3.0.1 Uncertainty-aware segmentation prototypes (extension)

A point prototype can be unstable when $K_m = 1$ or when the support images are not representative of the query domain. As a designated extension, a channel-wise support variance may be estimated:

$$v_c^l = \frac{1}{K_m}\sum_{i=1}^{K_m}\left(p_{c,i}^l - p_c^l\right)^2, \qquad c \in \{f, b\}, \tag{S4}$$

where $p_{c,i}^l$ is the prototype from support image i, stabilised by a source-validation shrinkage coefficient ρ:

$$\tilde{v}_c^l = (1-\rho)v_c^l + \rho\,\text{mean}(v_c^l), \qquad 0 \le \rho \le 1, \tag{S5}$$

with the uncertainty-adjusted similarity

$$\tilde{s}_c^l(x_q, u) = -\frac{\left(F^l(x_q, u) - p_c^l\right)^2}{\tilde{v}_c^l + \epsilon}, \tag{S6}$$

with channel-wise division and aggregation across channels. This variant is not part of the executed pipeline: the point-prototype model is the primary and only reported configuration. We retain the variance-weighted form here only as a pre-specified extension.

S3.3 Dense segmentation metric definitions (referenced from main text, 'Evaluation metrics').

Dice and intersection-over-union (IoU) are computed per case as:

$$\text{Dice} = \frac{2|M \cap \hat{M}|}{|M| + |\hat{M}|}, \qquad \text{IoU} = \frac{|M \cap \hat{M}|}{|M \cup \hat{M}|}. \tag{S7}$$

S4 Pre-registered protocols: full specifications

S4.1 Phase-4 joint fine-tuning: full regimen (referenced from main text, 'Pre-registered joint partial-supervision phase'; the v1 gate failure and the 0.8066 guard are retained in the main text).

Training follows protocol v2 (amended 2026-08-21, before any family-G/H evaluation). *Phase W* warms the classification head on the *frozen* trunk (AdamW 10^{-4} / 10^{-4}, cosine annealing, 25 epochs, classification batches only), placing the head near its conditional minimum before the trunk is touched. *Phase J* then unfreezes the trunk and jointly fine-tunes on the interleaved

batches at 10^{-5}, an order of magnitude below the ERM regimen, with cosine annealing, at most 15 epochs, patience 4, seed 42. Checkpoint selection maximises source validation AUROC *subject to a segmentation guard*: TN3K-validation Dice ≥ 0.8066 (the pre-registered non-collapse gate constant plus a 0.01 margin); the test split never participates in selection. The guard exists because protocol v1 failed this gate. Under a single-stage joint fine-tune at 10^{-4} with AUROC-only selection, the classification gradient of the freshly initialised head rotated the shared encoder from the first epoch: saved-model TN3K-test Dice was 0.6686 against a gate of 0.7966. The harness itself was validated by a pristine-checkpoint control that reproduced the locked reference exactly. Should no joint epoch satisfy the guard, the best-Dice state is retained for diagnosis and the non-collapse gate is allowed to fail. The joint phase is then reported as a negative result rather than analysed. Both arms share identical batch/augmentation streams (`set_seed(42)` first, warmup consuming a fixed, data-independent number of draws per arm) and differ only in the classification pooling region. Evaluation re-uses the locked caches wherever the protocol permits. The in-domain family G (pre-declared, Bonferroni within family) comprises: G1 JOINTg − ERM AUROC (is the in-domain fine-tuning gap closed?), G2 JOINTg − R1p (frozen versus joint grounding), G3 JOINTg − JOINT-img (grounding versus multi-tasking), G4 sensitivity@spec90 JOINTg − ERM (raw-raw). The joint model then enters the frozen external battery defined by the main-text transfer protocol, temperature fitted on its own TN5000 validation probabilities only. The TNUI-family operating-point behaviour (pre-specified as primary) and the DDTI descriptive rows are reported exactly as for earlier models. Acceptance gates fixed here: the TN5000 manifest must re-derive the locked split sizes; the joint segmentation branch must not collapse relative to the locked Stage-1 references; and the ERM sanity gate against the locked ERM probabilities applies unchanged. A pre-registered uncertainty component derives predictive entropy and probability margins from the calibrated probabilities and reports coverage-versus-malignant-FNR selective-prediction curves on the source test and the pre-declared pool.

S4.2 Phase-5 audit gate: full A1–A5 mechanics AND the verbatim pre-registered family J definition (referenced from main text, 'Case-level external validation on ThyUS2Path'; gate thresholds, the WITHHELD rule, the amendment date, and the J1 primary contrast are retained in the main text).

Audit-gated inference (A1–A5). No metric cell may run before the audit gate, and no inferential statement is unsealed before the licence. The checks execute in order:

A1 label coverage. A two-table per-batch prefix join maps each image identifier (first underscore-delimited token, namespaced by batch to forestall cross-batch collisions) onto its batch's label table (`histo_label` $\in \{0, 1\}$). Per-batch coverage must reach 95%, and the surviving join is cross-checked against the corpus's own image map.

A2 case assembly. A case is the set of images sharing its batch-namespaced patient key. The primary analytic unit is the case, scored by the unweighted mean of its per-image calibrated probabilities; a first-listed-image view is pre-declared as a descriptive sensitivity analysis.

A3 overlap screening. SHA-256 exact hashing plus perceptual-hash near-duplicate detection against every support or training pool (TN5000, TNUI-2021, the TN3K training pool); a single matching image excludes the *whole* case (conservative policy).

A4 cohort gates. Minimum assembled size (300 cases) and a malignancy-prevalence window of $[0.20, 0.85]$.

A5 polarity attestation. The operator inspects a seeded grid of benign- and malignant-labelled images against the corpus documentation and attests that label polarity matches ultrasonic appearance; only then is the inference licence set.

The pre-registered family is *computed* under the sealed discipline but reported as WITHHELD (descriptive statistics only) unless A1–A4 pass computationally *and* A5 is attested. The licence upgrade flips the significance column only and recomputes nothing. The amendment fixing the two-table join, the case-mean primary unit, and the conservative whole-case exclusion policy was adopted 2026-08-22, strictly before any target-side forward pass.

Pre-registered family J (case level, 5,000 paired bootstrap resamples, two-sided empirical p, Bonferroni $\alpha = 0.05/4 = 0.0125$; raw-raw convention; thresholds and temperatures from TN5000 validation only, nothing refitted): **J1** sensitivity@spec90 R1p − ERM, the primary external operating-point replication, with McNemar discordants on malignant cases; **J2** AUROC R1p − ERM (rank transfer, secondary); **J3** sensitivity@spec90 JOINT-g − R1p (external analogue of H2); **J4** AUROC R1p − R0 (grounding versus whole image). Oracle regimes (R1g/R2g) are excluded by pre-declaration because ThyUS2Path provides no lesion annotation.
S4.3 Phase-6 explainability battery: full axis specifications, perturbation recipes, cohort paragraph, and the verbatim pre-registered family K definition (referenced from main text, 'Pre-registered lesion-grounded explainability battery'; axis summaries, the ELA reproduction assertion, the cascade self-gate constant, cohort sizes, and the K1 primary contrast are retained in the main text).

Axes. (a) *Plausibility*: explanation concentration in the TN5000 expert box — the *localisation ratio* = in-box explanation share divided by box-area share (segmentation-region precision for descriptor heads; CAM energy for ERM), plus descriptive pointing-game hits. The ThyUS2Path side is descriptive only (predicted-mask agreement; no GT). (b) *Faithfulness*: deletion (primary) and insertion curves under a per-image Gaussian-blur baseline ($\sigma = 8$, 8 steps), raw probabilities (raw-raw convention), masks held fixed at the original prediction (isolated descriptor sensitivity), and one seeded random pixel ordering per image as control. The *faithfulness gap* is the trapezoidal area difference between the random-ordering and attribution-guided curves. (c) *Stability*: seeded Gaussian noise ($\sigma = 0.02$) and affine jitter ($\pm 3°$, ± 5 px) with the full pipeline (masks) recomputed; Spearman correlation of the clean and perturbed CAMs (affine maps inverse-warped before comparison). (d) *Randomisation sanity*: cascading reinitialisation — R1p head weights; ERM fc → fc+layer4 → fc+layer4+layer3 — on a seeded 160-image probe; the battery self-gates: mean $|\rho| < 0.2$ at the final cascade stage, asserted.

Cohorts. TN5000-test (canonical Phase-3 chain, n=950, 693 malignant; expert boxes as GT) and a seeded ThyUS2Path case-first-image panel (n=600, drawn from the licensed Phase-5 839-case manifest). Metrics on this panel are label-free, so the battery is independent of the Phase-5 licence.

Pre-registered family K (image level; 5,000 paired bootstrap resamples; two-sided empirical p; Bonferroni $\alpha = 0.05/4 = 0.0125$; seed 42): **K1** GT-localisation ratio R1p − ERM on TN5000-test (PRIMARY plausibility); **K2** deletion faithfulness gap R1p − ERM (TN5000-test); **K3** deletion faithfulness gap R1p − ERM (ThyUS2Path panel; label-free); **K4** deletion faithfulness gap JOINT-g − JOINT-img (TN5000-test; explanation-space mirror of G3). Everything else — insertion curves, pointing games, CAM-variant robustness, stability, cascade curves — is descriptive with pre-declared expectations.
S4.4 External classification transfer protocol: full gate specifications, tolerances, replication gate, and pooling/power policy (referenced from main text, 'External classification transfer protocol'; the two-sentence gate-logic summary is retained in the main text).

Frozen objects. Encoder, fusion decoder, fixed support episode (identifier-asserted against the archive), descriptor code, head weights, operating thresholds, and temperatures are byte-identical to the source phase.

Tolerance gates (checked before any external table).

1. **Support-episode gate:** episode identifiers must equal the archived identifiers (seed-drift assert).
2. **Head-provenance gate:** heads are loaded from the locked phase caches. Where a head must be retrained from cached features under the identical protocol (the random-number generator, RNG, is not exactly reproducible), its source validation AUROC must lie within ± 0.01 of the locked value; otherwise the run stops.
3. **ERM sanity gate:** the baseline's source-test AUROC, recomputed from the archived probabilities/checkpoint, must match the locked value within ± 0.002.
4. **Segmentation-QC gate:** external predicted-mask Dice is reported against the locked Phase-2 zero-shot references before any classification analysis.

Replication gate (governs pooling). When a previously locked external cohort is recomputed in a later session, the new table must match the locked table within absolute tolerance (0.0015 with cached heads; 0.01 with tolerated retrains) on every row. If any row drifts beyond tolerance, all analyses depending on pooling are *withheld* (flagged `POOLED_OK=False`; the sections report withholding rather than crashing or silently recomputing).

Pooling policy. Exactly one pool is pre-declared: TNUI-2021 test + validation, with its own test family (P1–P5, per-family Bonferroni). Later cohorts (TNUI-2021 train, DDTI) are analysed as individual families (E, F) with their own Bonferroni screens. Cross-cohort evidence matrices (AUROC and operating-point sensitivity across all cohorts) are descriptive only and carry no inferential claims beyond the pre-declared families.

Power policy. External cohorts with very few benign cases yield wide intervals; the pre-declared remedies are direction-consistency reporting across cohorts and the single pooled analysis. Underpowered individual tests are labelled as such rather than narrated as effects.

S5 Reproducibility artefact manifest

Referenced from main text, 'Reproducibility and implementation controls'.

Released or documented:

- dataset versions and access locations;
- clean manifests re-derived from raw files and hash-verified against the audit manifests;
- duplicate-drop lists;
- the TN5000 manifest with locked split sizes and per-split prevalence;
- preprocessing and augmentation configurations;
- episode manifests, including the archived fixed support episode;
- feature-bank caches with quality-control fields;
- cached model probabilities for every reported table;
- label parsers and the DDTI vote profile;
- the TNUI label-polarity record (assumption-with-evidence);

- locked result tables cited by later phases as tolerance references;
- the audit-hash manifest of every external phase;
- seeds and software versions;
- model checkpoints (the Stage 1 reference checkpoint and the fusion decoder are versioned with the repository artefacts); and
- scripts for metrics and statistics.

S6 Algorithmic summary

Algorithm S1, referenced from main text, 'Algorithmic summary'.

Algorithm S1 Lesion-Grounded Few-Shot Segmentation–Classification Framework

Require: Source segmentation domain with masks; pathology-labelled source classification domain; external cohorts; budgets K_m, K_c; query counts; seeds

Ensure: Segmentation masks; malignancy predictions; calibration and operating points; explanations

1: Hash and de-duplicate images; rebuild and verify manifests; group by patient/nodule where identifiers exist.
2: **Stage 1:** Train the supervised ResNet34-UNet reference on the source training pool (Tversky objective, EMA, val-only threshold sweep, frozen threshold).
3: **Stage 2:** Freeze the encoder; train the fusion decoder episodically on source episodes with nested budgets and joint augmentation.
4: Compute multi-scale fg/bg prototypes (the main-text multi-scale foreground/background prototype equations); predict query masks via the main-text query-mask decoder equation or the direct variant
5: Evaluate in-domain, zero-shot, and mask-shot segmentation over 200 episodes each, all budgets on matched queries; bootstrap episode means.
6: **Stage 3:** Archive the fixed support episode; segment all classification cohorts zero-shot; build and cache the five descriptor regimes; train linear heads, the prototype budget sweep, and the ERM baseline on the source domain only.
7: **Stage 4:** Freeze per-model temperature and the specificity-0.90 operating threshold from source validation (raw-raw convention).
8: **Stage 5:** Apply the frozen pipeline to each external cohort; enforce tolerance, sanity, and replication gates; run the pre-declared paired families and the single pre-declared pool.
9: **Phase 4 (pre-registered):** joint partial-supervision fine-tune; family G in-domain; external battery under the same gates.
10: Evaluate lesion alignment, faithfulness, stability, and randomisation of explanations on internal and external cases.

S7 Supplementary results

This section preserves, verbatim, the per-cohort and operational tables displaced from the 16-page main text, and – in Tables S5 and S22 – the secondary and tertiary pre-registered contrasts relocated row-for-row from the main-text family tables. Each block notes the main-text subsection that cites it.

Table S1, referenced from main text, 'Cohorts, audits, and frozen operating conventions'.
Table S2, referenced from main text, 'Supervised segmentation reference and external gap'.
Table S3, referenced from main text, 'Few-shot support-conditioned segmentation'.

Table S1: Frozen-pipeline segmentation QC in classification phases (locked, Phase-1/2 pipeline at the fixed K_m=20 support episode; Dice vs cohort masks at the locked 0.5 binarisation). —: not available — predicted-mask area was not recorded by the Phase-3d/3e banks (Dice/empty only), and the TNUI-train empty-rate line was not archived; descriptive QC only, both recoverable on re-run.

Cohort	mean Dice	empty-mask rate	mean mask area
TN5000 (all splits)	n/a (expert boxes; no pixel masks)	1.1% val / 1.2% test	0.037–0.038
TNUI-2021 test	0.7046	0.0%	0.033
TNUI-2021 val	0.7307	1.5%	—
TNUI-2021 train	0.7241	—	—
DDTI (labelled subset)	0.6132	3.7%	—

Table S2: Phase-1 supervised segmentation reference (locked; from `aggregate_metrics.csv`). 95% case-bootstrap CIs for Dice; HD95 is the median over cases with both masks non-empty (n given; letterboxed 384^2 pixels). TNUI per-split TTA Dice: train 0.7894, val 0.8029.

Set	TTA	Dice (95% CI)	IoU	Sens	Prec	HD95 med (n)
TN3K test (n=614)	yes	0.8466 (0.8336–0.8589)	0.7603	0.9108	0.8240	11.16 (613)
	no	0.8441 (0.8310–0.8565)	0.7572	0.9149	0.8165	11.31 (613)
TNUI-2021 all (n=1371)	yes	0.7877 (0.7778–0.7972)	0.6791	0.9117	0.7329	9.25 (1366)
	no	0.7766 (0.7663–0.7867)	0.6665	0.9150	0.7160	10.00 (1368)
TNUI-2021 test (n=138)	yes	0.7464 (0.7098–0.7805)	0.6315	0.9048	0.6837	10.63 (138)
	no	0.7367 (0.6977–0.7713)	0.6211	0.9072	0.6699	11.01 (138)
DDTI (n=635)	yes	0.7396 (0.7213–0.7561)	0.6288	0.8324	0.7327	22.00 (629)
	no	0.7431 (0.7258–0.7589)	0.6301	0.8471	0.7252	23.22 (629)

Table S3: Phase-2 direct (parameter-free) variant, query Dice (locked). IoU for the decoder at K=20: 0.6873 / 0.5808 / 0.5471 / 0.5598 / 0.5602 across the five conditions in table order.

Condition	K=1	K=5	K=10	K=20
TN3K test (in-domain)	0.6205 (0.5987–0.6432)	0.6778 (0.6642–0.6911)	0.6788 (0.6656–0.6922)	0.6751 (0.6626–0.6877)
TNUI test (zero-shot)	0.4744 (0.4510–0.4975)	0.4943 (0.4727–0.5161)	0.4964 (0.4762–0.5170)	0.4985 (0.4795–0.5174)
TNUI test (K-shot)	0.4906 (0.4657–0.5144)	0.5965 (0.5823–0.6104)	0.6175 (0.6040–0.6297)	0.6277 (0.6156–0.6399)
DDTI (zero-shot)	0.5111 (0.4888–0.5332)	0.5637 (0.5480–0.5787)	0.5665 (0.5519–0.5800)	0.5682 (0.5544–0.5813)
DDTI (K-shot target)	0.5789 (0.5616–0.5965)	0.6429 (0.6323–0.6545)	0.6507 (0.6398–0.6624)	0.6529 (0.6418–0.6642)

Table S4: TN5000 test classification (locked; n=950, malignant 693). ECE 15-bin; Brier and ECE-post use validation-fitted temperature; sens@spec90 is evaluated at the frozen raw validation threshold (raw-raw convention). CIs: 95% case bootstrap (2,000 replicates).

Model	AUROC (95% CI)	AUPRC	sens@spec90	Brier	ECE pre/post	T_{val}
ERM whole-image (ResNet34 ft)	0.9401 (0.9232–0.9553)	0.9738	0.8023	0.0878	0.0893/0.0634	1.969
R0 image — linear	0.8534 (0.8271–0.8798)	0.9358	0.6479	0.1520	0.1278/0.1244	0.900
R1p lesion — linear	0.8902 (0.8670–0.9122)	0.9519	0.7273	0.1515	0.1788/0.1734	0.847
R2p lesion+ring — linear	0.8829 (0.8584–0.9061)	0.9464	0.6869	0.1538	0.1721/0.1705	0.956
R1g GT box — linear (oracle)	0.9105 (0.8899–0.9299)	0.9635	0.8254	0.1260	0.1339/0.1302	0.927
R2g GT box+ring — linear (oracle)	0.9029 (0.8807–0.9239)	0.9569	0.7763	0.1449	0.1750/0.1721	0.927

Table S4, referenced from main text, 'In-domain classification with frozen descriptors'.
Table S5: secondary and tertiary pre-registered contrasts of families T, G, H, and J, relocated row-for-row from main-text Tables 'Phase-3b paired tests', 'Phase-4 pre-registered families', and 'Phase-5 pre-registered family J' (referenced from the corresponding main-text subsections). Cohort, resample count, and Bonferroni bar per family follow each family header row; conventions as in the main-text tables ($\Delta = A-B$; empirical p floored at $p<0.0004$).
Table S6, referenced from main text, 'In-domain classification with frozen descriptors'.

Table S5: Supplementary secondary/tertiary pre-registered contrasts (relocated rows; locked values). Family T: TN5000 test, 5,000 paired bootstrap resamples, α=0.0083. Family G: in-domain TN5000 test, 5,000 resamples, α=0.0125. Family H: pooled TNUI test+val (n=413, 392 positives), 2,000 resamples, α=0.0167. Family J: ThyUS2Path case level (n=839, 522 malignant), 5,000 resamples, α=0.0125.

Test	Contrast	Metric	Δ (95% CI)	p	Sig.	Question
Family T (secondaries), in-domain TN5000 test						
T3	R1g − R1p	AUROC	+0.0202 (0.0056, 0.0349)	0.0048	YES	seg-error cost ≈2 pts
T4	R1p − R2p	AUROC	+0.0074 (−0.0023, 0.0176)	0.1388	no	ring adds nothing (H6 null)
T5	R1p − ERM	sens@spec90	−0.0749 (−0.1126, −0.0390)	<0.0004	YES	ERM wins op. point
T6	R1g − R1p	sens@spec90	+0.0982 (0.0693, 0.1270)	<0.0004	YES	oracle > predicted ROI
McNemar (T5): ERM-right/R1p-wrong 114 vs R1p-right/ERM-wrong 62, $\chi^2 = 14.78$.						
Family G (G2–G4), in-domain						
G2	JOINT-g − R1p	AUROC	−0.0205 (−0.0365, −0.0045)	0.0124	YES	joint vs frozen grounding — frozen wins
G3	JOINT-g − JOINT-img	AUROC	+0.1489 (0.1136, 0.1848)	<0.0004	YES	grounding beats multi-tasking
G4	JOINT-g − ERM	sens@spec90	−0.2376 (−0.2785, −0.1962)	<0.0004	YES	in-domain operating point — ERM wins
McNemar (G4, test positives): JOINT-g-right/ERM-wrong 41 vs ERM-right/JOINT-g-wrong 206, $\chi^2 = 108.89$ ($p \ll 0.01$) — ≈5:1 discordants against JOINT-g at the frozen point.						
Family H (H2–H3), pooled TNUI test+val (external)						
H2	JOINT-g − R1p	sens@spec90 (pooled)	−0.0534 (−0.0969, −0.0102)	0.011	YES	frozen grounding transfers better
H3	JOINT-g − ERM	AUROC (pooled)	−0.0109 (−0.1442, 0.1322)	0.858	no	external rank (secondary) — null
Family J (J2–J4), ThyUS2Path case level (external)						
J2	R1p − ERM	AUROC	−0.0401 (−0.0801, −0.0015)	0.0436	no	rank transfer — null at family bar
J3	JOINT-g − R1p	sens@spec90	−0.0978 (−0.1264, −0.0709)	<0.0004	YES	frozen grounding transfers better (H2 analogue)
J4	R1p − R0	AUROC	−0.0322 (−0.0709, 0.0075)	0.1104	no	grounding vs whole image — null

Table S6: Frozen operating points (locked): per-model thresholds chosen on raw validation probabilities at specificity ≥ 0.90, then applied unchanged to the test split (raw-raw convention). The val→test sensitivity transfer gap is smallest for the oracle and grounded heads (R1g +0.60 pts, R1p −1.71 pts) and largest for ERM (−5.33 pts), foreshadowing the external operating-point behaviour.

Model	threshold (frozen)	val spec	val sens	test sens@spec90
ERM	0.9233	0.9052	0.8556	0.8023
R0 image	0.6708	0.9052	0.6417	0.6479
R1p lesion	0.5323	0.9052	0.7444	0.7273
R2p lesion+ring	0.5789	0.9138	0.7278	0.6869
R1g GT box (oracle)	0.5194	0.9052	0.8194	0.8254
R2g GT box+ring (oracle)	0.4699	0.9052	0.8056	0.7763

Table S7, referenced from main text, 'In-domain classification with frozen descriptors' (error-propagation quartiles).

Table S7: Error propagation (RQ4, locked; TN5000 test, n=950): oracle-minus-predicted AUROC gap within predicted-mask-confidence quartiles (237–238 cases each). gap = AUROC(GT-ROI) − AUROC(pred-ROI); positive = oracle better. The gap shows no monotone confidence relationship and is descriptively largest at the highest confidence (single split; no per-quartile CIs).

Confidence quartile	n	AUROC pred-ROI	AUROC GT-ROI	gap
Q1 (low)	238	0.8871	0.9081	+0.0209
Q2	237	0.8720	0.8834	+0.0114
Q3	237	0.8811	0.8769	−0.0042
Q4 (high)	238	0.8433	0.9022	+0.0589

Table S8, referenced from main text, 'In-domain classification with frozen descriptors' (label-budget sweep).
Table S9, referenced from main text, 'External classification battery' (TNUI-2021 test; per-cohort family Tc).
Table S10, referenced from main text, 'External classification battery' (TNUI-2021 val and the

Table S8: Class-prototype label-budget sweep (locked; TN5000 test). Cosine-margin prototype classifier on frozen descriptors; AUROC mean ± SD (SD to 3 d.p.) over 25 independent per-class label draws at each budget K_c; the *all* row uses the full training split once (single draw). Grounded heads dominate the image budget at every K and saturate by K=50; R0 does not.

Budget K_c	R0 image	R1p lesion	R2p lesion+ring	R1g GT box	R2g GT box+ring
5	0.5058±0.050	0.5862±0.048	0.5778±0.044	0.6119±0.051	0.6066±0.033
10	0.5307±0.045	0.6190±0.043	0.6058±0.036	0.6477±0.053	0.6290±0.030
20	0.5398±0.044	0.6362±0.043	0.6246±0.037	0.6618±0.043	0.6484±0.032
50	0.5586±0.034	0.6383±0.037	0.6335±0.033	0.6570±0.033	0.6478±0.024
200	0.5827±0.032	0.6348±0.023	0.6294±0.023	0.6491±0.021	0.6441±0.015
all	0.5917	0.6379	0.6313	0.6508	0.6462

Support-sampling variance shrinks with budget ($\sigma \approx 0.05$ at K=5 falling to ≈ 0.02 at K=200). Absolute levels sit far below the linear heads' (0.59–0.66 vs 0.85–0.91), as expected for a cosine prototype rule on frozen descriptors.

Table S9: Phase-3c TNUI-test external classification (locked; n=139, 134 malignant). Thresholds frozen from TN5000-val raw probabilities (raw-raw); Brier/ECE-post use source-val-fitted temperatures T (frozen descriptors, tolerance-gated retrained heads). AUPRC is near-uninformative at prevalence 0.964. External ECE is poor for *every* model (0.24–0.43) — prevalence-shift miscalibration. Shift = in-domain minus external AUROC.

Model	AUROC (95% CI)	AUPRC	sens@spec90	Brier	ECE pre/post	$T_{\text{src val}}$	shift
ERM	0.5172 (0.3296–0.6937)	0.9745	0.5373	0.1963	0.2427/0.2674	1.969	0.423
R0	0.5791 (0.3082–0.8154)	0.9746	0.4104	0.3074	0.4318/0.4244	0.927	0.273
R1p	0.7119 (0.5612–0.8768)	0.9871	0.6940	0.2036	0.3173/0.3060	0.847	0.177
R2p	0.6776 (0.3630–0.9058)	0.9797	0.6493	0.2157	0.3377/0.3320	0.927	0.207
R1g (true mask)	0.8343 (0.7269–0.9270)	0.9933	0.6716	0.2071	0.3183/0.3145	0.927	0.075
R2g (mask+ring)	0.8507 (0.7500–0.9448)	0.9940	0.5000	0.2828	0.4175/0.4173	0.985	0.052

Paired (α=0.01; n_{boot}=4952 for AUROC after discarding single-class resamples, 5000 for sens): T1c +0.1315 (−0.1208, 0.4187; p=0.324) underpowered-null; T2c −0.1940 (−0.4193, 0.0651; p=0.117) sign-flipped vs in-domain T2, CI crosses 0; T3c +0.1230 (0.0000, 0.2896; p=0.049) nominal only; T4c −0.3170 (−0.4762, −0.1423; p=0.0004) **YES**; T5c −0.1570 (−0.2537, −0.0597; p=0.0028) **YES**.

pre-declared pool; families Td and P).
Table S11, referenced from main text, 'External classification battery' (DDTI and TNUI-2021 train; families F and E).
Figure S1, referenced from main text, 'External classification battery' (external verdict paragraph).

Table S10: Phase-3d: TNUI-val (n=274; 258 malignant/16 benign) and the pre-declared pooled TNUI test+val (n=413; 392/21). Locked thresholds from TN5000-val (raw-raw). Val-panel ECE-pre range 0.24–0.42; pooled panel reports post-temperature ECE. Replication gate vs locked 3c: zero drift.

Model	AUROC (95% CI)	AUPRC	sens@spec90	Brier	ECE post
TNUI-2021 val					
ERM	0.6237 (0.4765–0.7670)	0.9617	0.5504	0.1879	0.2497
R0	0.5901 (0.3824–0.7877)	0.9338	0.4031	0.3019	0.4162
R1p	0.5775 (0.3895–0.7484)	0.9422	0.6512	0.2434	0.3278
R2p	0.6008 (0.4193–0.7649)	0.9520	0.6318	0.2196	0.3121
R1g (true mask)	0.6126 (0.4516–0.7640)	0.9557	0.6434	0.2475	0.3112
R2g (mask+ring)	0.5770 (0.4111–0.7370)	0.9536	0.5271	0.3059	0.3902
Pooled TNUI test + val (single pre-declared pool)					
ERM	0.5969 (0.4756–0.7106)	0.9660	0.5459	0.1907	0.2557
R0	0.5850 (0.4284–0.7370)	0.9505	0.4056	0.3038	0.4172
R1p	0.6079 (0.4726–0.7385)	0.9564	0.6658	0.2300	0.3156
R2p	0.6144 (0.4725–0.7519)	0.9615	0.6378	0.2183	0.3188
R1g (true mask)	0.6635 (0.5414–0.7780)	0.9677	0.6531	0.2339	0.3111
R2g (mask+ring)	0.6340 (0.5031–0.7591)	0.9661	0.5179	0.2981	0.3953

Pooled paired family (α=0.01, 5000 resamples): P1 R1p−R0 +0.0227 (−0.0718, 0.1234; p=0.662) no; P2 ERM−R1p −0.0118 (−0.1264, 0.0991; p=0.852) no; P3 R1g−R1p +0.0554 (−0.0077, 0.1293; p=0.096) no; P4 ERM−R1g −0.0672 (−0.1822, 0.0382; p=0.224) no; P5 ERM−R1p sens −0.1201 (−0.1811, −0.0612; p<0.0004) **YES** — the 3c operating-point inversion replicates across cohorts.

Table S11: Phase-3e arms (locked): DDTI XML-labelled subset (n=189; 145/44) and TNUI-train (n=966; 915/51). Conventions as in Table S10. DDTI is at chance for every model, with true-mask oracle rows *below* 0.5 — burned-in acquisition annotations are a documented confound.

Model	AUROC (95% CI)	AUPRC	sens@spec90	Brier	ECE post
DDTI (independent institution)					
ERM	0.4746 (0.3729–0.5740)	0.7675	0.4828	0.2704	0.2699
R0	0.5116 (0.4196–0.6034)	0.7923	0.1724	0.4174	0.4364
R1p	0.4967 (0.4059–0.5844)	0.7874	0.3379	0.3874	0.3864
R2p	0.5017 (0.4057–0.5909)	0.7928	0.3034	0.3696	0.3559
R1g (true mask)	0.4216 (0.3315–0.5066)	0.7668	0.4276	0.3940	0.3927
R2g (mask+ring)	0.4143 (0.3210–0.5020)	0.7435	0.2759	0.4334	0.4263
TNUI-train (heterogeneity probe)					
ERM	0.5936 (0.5140–0.6706)	0.9604	0.5519	0.1987	0.2593
R0	0.5472 (0.4556–0.6407)	0.9505	0.4230	0.3104	0.4108
R1p	0.6316 (0.5491–0.7160)	0.9633	0.6776	0.2151	0.2968
R2p	0.6441 (0.5714–0.7170)	0.9698	0.6404	0.2136	0.2978
R1g (true mask)	0.6459 (0.5603–0.7301)	0.9643	0.6656	0.2248	0.3036
R2g (mask+ring)	0.6078 (0.5222–0.6953)	0.9605	0.5148	0.3011	0.3940

E family (DDTI, α=0.01, 5000 resamples): E1 R1p−R0 −0.0162 (−0.1064, 0.0768; p=0.724); E2 ERM−R1p −0.0215 (−0.1220, 0.0823; p=0.676); E3 R1g−R1p −0.0749 (−0.1498, −0.0006; p=0.048) nominal only — oracle *worse* than predicted ROI on DDTI; E4 ERM−R1g +0.0534 (−0.0407, 0.1473; p=0.273); **E5 ERM−R1p sens +0.1456 (0.0621, 0.2345;** p=0.0012**) YES** — operating-point inversion on the independent cohort.
F family (TNUI-train): F1 R1p−R0 +0.0846 (0.0083, 0.1642; p=0.025) nominal-positive; F2 −0.0381 (p=0.326); F3 +0.0144 (p=0.566); F4 −0.0525 (p=0.222); **F5 ERM−R1p sens −0.1259 (−0.1628, −0.0885;** p<0.0004**) YES** — the grounded operating-point advantage replicates in the third TNUI cohort.

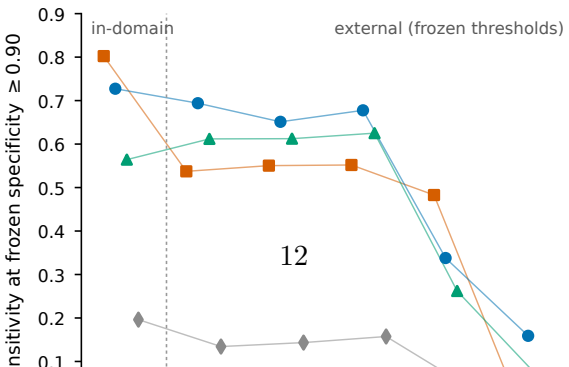


Table S12, referenced from main text, 'Joint partial-supervision fine-tuning' (in-domain levels).

Table S12: Phase-4 in-domain TN5000 test (n=950; locked, protocol v2). Columns as Table S4; JOINT-row CIs from this phase's 2,000-replicate case bootstrap. Both segmentation non-collapse gates passed (TN3K-test Dice 0.8258 JOINT-g / 0.8165 JOINT-img against the ≥ 0.7966 gate). Context rows are the locked Phase-3 values (Table S4).

Model	AUROC (95% CI)	AUPRC	sens@spec90	Brier	ECE pre/post	T_{val}
JOINT-g	0.8698 (0.8426–0.8939)	0.9423	0.5642	0.1573	0.1971/0.1682	0.523
JOINT-img	0.7207 (0.6824–0.7591)	0.8460	0.1962	0.2103	0.2241/0.1927	0.200
Locked Phase-3 context rows						
ERM whole-image (ResNet34 ft)	0.9401 (0.9232–0.9553)	0.9738	0.8023	0.0878	0.0893/0.0634	1.969
R0 image — linear	0.8534 (0.8271–0.8798)	0.9358	0.6479	0.1520	0.1278/0.1244	0.900
R1p lesion — linear	0.8902 (0.8670–0.9122)	0.9519	0.7273	0.1515	0.1788/0.1734	0.847
R2p lesion+ring — linear	0.8829 (0.8584–0.9061)	0.9464	0.6869	0.1538	0.1721/0.1705	0.956
R1g GT box — linear (oracle)	0.9105 (0.8899–0.9299)	0.9635	0.8254	0.1260	0.1339/0.1302	0.927
R2g GT box+ring — linear (oracle)	0.9029 (0.8807–0.9239)	0.9569	0.7763	0.1449	0.1750/0.1721	0.927

Table S13, referenced from main text, 'Joint partial-supervision fine-tuning' (external levels).

Table S13: Phase-4 external battery (locked; thresholds from each model's own TN5000-val raw probabilities, raw-raw convention). Segmentation QC for the JOINT-g trunk: external Dice 0.7201 (TNUI-test), 0.7651 (TNUI-val), 0.7543 (TNUI-train), 0.6738 (DDTI; TN3K gate 0.8258); empty-mask rate $\leq 0.3\%$ on every cohort. Frozen rows (ERM, R1p) replicate the locked Phase-3c/3d/3e external values exactly. †descriptive cohorts under the locked pooling policy (inference only within the pooled family, family H, the main-text Phase-4 family-G/H table).

		AUROC				sens@spec90			
Cohort	n (mal)	ERM	JOINT-g	JOINT-img	R1p	ERM	JOINT-g	JOINT-img	R1p
TNUI-test	139 (134)	0.5172	0.7075	0.6493	0.7119	0.5373	0.6119	0.1343	0.6940
TNUI-val	274 (258)	0.6237	0.5523	0.5337	0.5775	0.5504	0.6124	0.1434	0.6512
TNUI-train†	966 (915)	0.5936	0.5845	0.5270	0.6316	0.5519	0.6251	0.1574	0.6776
DDTI†	189 (145)	0.4746	0.5210	0.5296	0.4967	0.4828	0.2621	0.0414	0.3379
Derived pooled (test+val) operating points: sens@spec90 R1p $\approx 0.666 >$ JOINT-g $\approx 0.612 >$ ERM ≈ 0.546.									

Table S14, referenced from main text, 'Joint partial-supervision fine-tuning' (selective-prediction rows).
Figure S2, referenced from main text, 'Joint partial-supervision fine-tuning' (selective-prediction paragraph).

Table S14: Phase-4 selective prediction (Gap 6; locked). Coverage = the most-confident-retained fraction at the frozen raw validation threshold (raw-raw; the validation-fitted temperature, T=0.523, supplies the confidence margin only — no threshold or temperature is re-estimated). *mal. FNR* = malignant false-negative rate among retained cases (equal to 1−sens@spec90 row-wise); *mal. share* = fraction of the cohort's malignant cases falling inside the retained confident set. Companion figure: `phase4_selective.png` (both curves monotone in coverage).

Coverage	n retained	mal. FNR	sens@spec90	mal. share retained
TN5000-test (in-domain, n=950, 693 malignant)				
100%	950	0.4358	0.5642	1.000
90%	855	0.3673	0.6327	0.8918
80%	760	0.2826	0.7174	0.7864
70%	665	0.1602	0.8398	0.6667
60%	570	0.1490	0.8510	0.5714
50%	475	0.1262	0.8738	0.4690
TNUI pooled (test+val; external, n=413, 392 malignant)				
100%	413	0.3878	0.6122	1.000
90%	372	0.3220	0.6780	0.9031
80%	330	0.2357	0.7643	0.8010
70%	289	0.1594	0.8406	0.7041
60%	248	0.1398	0.8602	0.6020
50%	206	0.1026	0.8974	0.4974

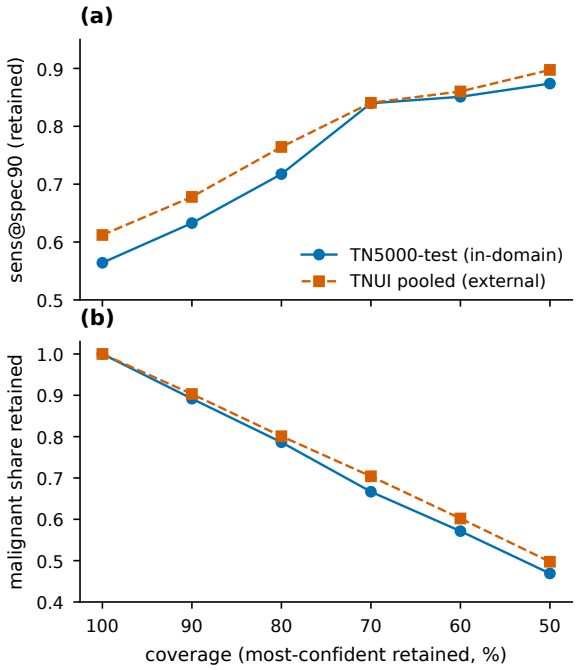


Figure S2: **Selective prediction is referral-shaped, in-domain and externally.** (a) Sensitivity at the frozen operating point as lower-confidence cases are referred (coverage = most-confident retained fraction; confidence from the validation-fitted temperature margin, no refitting). (b) Fraction of each cohort's malignant cases inside the retained set. At 50% coverage, sensitivity reaches 0.874 (TN5000-test) and 0.897 (pooled TNUI), and 86–87% of the frozen point's malignant misses fall into the referred half.

Table S15, referenced from main text, 'Case-level external validation on ThyUS2Path' (per-model case metrics).
Table S16, referenced from main text, 'Case-level external validation on ThyUS2Path' (domain-

Table S15: ThyUS2Path case-level external battery (locked; n=839 cases, 522 malignant; unit = case, score = mean of per-image calibrated probabilities; thresholds from each model's own TN5000-val raw probabilities, raw-raw convention). CIs: 2,000-replicate case bootstrap. Oracle regimes (R1g/R2g) not computable — ThyUS2Path provides no lesion annotation (pre-declared exclusion). Companion file: `phase5_main_table.csv`.

Model	AUROC (95% CI)	AUPRC	sens@spec90	Brier	ECE pre/post	T_{val}
ERM whole-image (ResNet34 ft)	0.5479 (0.5082–0.5868)	0.6662	0.0134	0.2873	0.2813/0.2145	1.969
R0 image — linear	0.5400 (0.4992–0.5791)	0.6408	0.0460	0.3486	0.3039/0.3139	0.927
R1p lesion — linear	0.5081 (0.4686–0.5494)	0.6300	0.1590	0.3423	0.2761/0.2942	0.847
R2p lesion+ring — linear	0.5211 (0.4805–0.5619)	0.6392	0.0881	0.3337	0.2771/0.2852	0.927
JOINT-g	0.5112 (0.4715–0.5506)	0.6342	0.0613	0.3200	0.2020/0.2607	0.523
JOINT-img	0.4780 (0.4387–0.5201)	0.6078	0.0153	0.3198	0.1506/0.2738	0.200

shift matrix).

Table S16: Domain shift (locked): TN5000-test AUROC (source) versus ThyUS2Path case-level AUROC (external). Source values are re-derived in the Phase-5 environment, not copied: ERM re-hits the locked 0.9401 sanity gate; R1p reproduces 0.8889 (within the 0.002 integrity tolerance of the locked 0.8902 reference); the JOINT rows match Table S12 exactly.

Model	AUROC TN5000-test	AUROC ThyUS2Path	Shift drop
ERM	0.9401	0.5479	0.3921
R0	0.8524	0.5400	0.3124
R1p	0.8889	0.5081	0.3808
R2p	0.8841	0.5211	0.3630
JOINT-g	0.8698	0.5112	0.3586
JOINT-img	0.7207	0.4780	0.2427
Derived mean shift drop: 0.3416 AUROC (range 0.2427–0.3921).			

Table S17, referenced from main text, 'Case-level external validation on ThyUS2Path' (first-listed-image view).

Table S17: First-listed-image sensitivity view (descriptive; locked). Each case is scored by its first-listed image only — identical frozen thresholds, no case aggregation — demonstrating that the collapse of Table S15 is not an artefact of the case-mean unit.

Model	AUROC (first image)	sens@spec90 (first image)
ERM	0.5328	0.1648
R0	0.5006	0.1475
R1p	0.4877	0.2375
R2p	0.4965	0.2069
JOINT-g	0.4813	0.1743
JOINT-img	0.4974	0.0632

Table S18, referenced from main text, 'Lesion-grounded explainability battery' (localisation ratios, in-box mass, entropy, pointing games).
Table S19, referenced from main text, 'Lesion-grounded explainability battery' (deletion/insertion areas incl. random-control mechanism).
Table S20, referenced from main text, 'Lesion-grounded explainability battery' (noise/affine stability rows).
Table S21, referenced from main text, 'Lesion-grounded explainability battery' (randomisation cascade stages).
Table S22: family-K secondary/tertiary contrasts, relocated row-for-row from the main-text family-

Table S18: Phase-6 plausibility against TN5000 expert boxes (n=950; all images boxed; localisation ratio = in-box share / box-area share; in-box mass = share of the ℓ_1-normalised explanation inside the box; hit = argmax inside box (pointing game, descriptive); entropy normalised to the 384^2 canvas. Score-CAM computed on the pre-registered seeded n=150 subsample (top-64 channels by $|\alpha|$). All quantities computed once, seed 42, no refit.

Explanation	loc. ratio mean (median)	in-box mass	pointing hit	norm. entropy
R1p-ELA (grounded, exact)	41.63 (27.83)	0.796	0.637	0.717
ERM Grad-CAM (layer4)	3.60 (4.44)	0.146	0.514	0.928
ERM Eigen-CAM	4.68 (4.80)	—	0.736	—
ERM Score-CAM (n=150)	3.57 (3.57)	—	0.580	—

Table S19: Phase-6 faithfulness on TN5000-test (deletion/insertion under per-image blur σ=8, 8 steps, raw-raw probabilities, masks fixed post-fallback; gap = trapezoid-area difference against the per-image seeded random-ordering control; insertion descriptive on the seeded n=300 subset). ThyUS-600 per-model curves are archived in `phase6_panel_thyus.npz`; the inferential external statement (K3) is in the main-text family-K table.

Model	del. AUC guided	del. AUC random	deletion gap	insertion gap (n=300)
R1p	0.3334	0.4986	0.1652	0.0351
ERM	0.3315	0.5564	0.2249	0.2043
JOINT-g	0.3886	0.4908	0.1022	0.0356
JOINT-img	0.4405	0.4815	0.0410	0.0144

Table S20: Phase-6 stability under perturbation (mean / median Spearman against the unperturbed explanation; seeded 160-image probes per cell; pre-declared descriptive, non-gated).

Cohort	Model	Perturbation	mean ρ	median ρ
TN5000-test	R1p	noise σ=0.02	0.9208	0.9972
TN5000-test	R1p	affine ($\pm3°$, ±5px)	0.7713	0.8815
TN5000-test	ERM	noise σ=0.02	0.8317	0.8865
TN5000-test	ERM	affine ($\pm3°$, ±5px)	0.8709	0.9245
ThyUS-600	R1p	noise σ=0.02	0.7990	0.9782
ThyUS-600	R1p	affine ($\pm3°$, ±5px)	0.6402	0.7909
ThyUS-600	ERM	noise σ=0.02	0.7545	0.7879
ThyUS-600	ERM	affine ($\pm3°$, ±5px)	0.8103	0.8574

Table S21: Phase-6 randomisation cascade (mean Spearman between the original and randomised explanations over the seeded 160-image TN5000 probe). Pre-registered self-gate $|\rho|<0.2$ at each model's final available stage: ERM **pass**, R1p **fail** — reported as failed. The fc-only row shows shallow readout randomisation cannot reach the bar for either model; follow-up diagnostics (decision rules committed pre-run): resampled-vs-resampled floor 0.0920 (elevation of the retained correlation +0.2072 above floor), per-feature attribution decorrelation complete (Spearman −0.0150). Originals verified untouched by checksum.

Stage	mean $\|\rho\|$	mean ρ	n
ERM fc	0.2948	−0.0618	160
ERM fc+layer4	0.1492	+0.1287	160
ERM fc+layer4+layer3 (final)	0.1205	−0.0135	160
R1p head weights (final)	0.2991	+0.2982	160

K table (referenced from main text, 'Lesion-grounded explainability battery'). Conventions as in

the main-text table (image-level paired bootstrap, 5,000 resamples, seed 42; Bonferroni α=0.0125*;* p=0.0 *typeset* p<0.0004*).*

Table S22: Supplementary family-K secondary contrasts (relocated rows; locked values). "YES (rev.)" = significant in the pre-registered-opposite direction (see main text).

Test	Metric	A	B	Δ(A−B) (95% CI)	Sig.	Note
K2	deletion gap	R1p	ERM	−0.0598 (−0.0761.. − 0.0437)	YES (rev.)	faithfulness, TN5000
K3	deletion gap	R1p	ERM	−0.0339 (−0.0511.. − 0.0171)	YES (rev.)	faithfulness, ThyUS-600
K4	deletion gap	JOINT-g	JOINT-img	+0.0613 (0.0560..0.0667)	YES	grounding, G3 mirror

S8 Narrative extracts preserved from the pre-compression main text

Sentences below are reproduced verbatim from the pre-compression main text wherever a reported detail has no other Supplementary home, so the compression removes nothing from the submission package.

From Material and Methods (architecture and preprocessing), verbatim. The working resolution is 384×384; the prototype module uses levels $\{e_2, e_3, e_4\}$ (strides 8/16/32; 128/256/512 channels) and descriptors pool $\{e_3, e_4\}$. Images are letterboxed; masks share the geometric transform (nearest-neighbour) and are binarised at grey level ≥ 127, selected by visual audit against > 0 and ≥ 250 alternatives; inputs are ImageNet-normalised. Score maps are bilinearly resized to the stride-8 grid (48×48) and stacked as $2L$ channels (fragment, equation region).

From Material and Methods (episode and budget mechanics), verbatim. Meta-training samples 300 episodes per epoch from the source pool, re-drawing K_m uniformly from $\{1, 5, 10, 20\}$ at every episode; validation uses 100 episodes and evaluation 200 per condition. Descriptors are L2-normalised, and per-class means over nested per-class draws from the training split ($K_c \in \{5, 10, 20, 50, 200, \text{all}\}$, 25 resamples per finite budget, fixed seed) are scored by the cosine margin, reported as mean ± standard deviation of test AUROC across draws; in-domain only.

From Material and Methods (decision conventions), verbatim. Four conventions were fixed after the paired-analysis audit: (1) thresholds and temperatures come from the source validation split only, frozen before any test or external evaluation; (2) the raw-raw convention never mixes raw-threshold binary decisions with calibrated probabilities (rank metrics are convention-invariant); (3) confidence intervals are case-level bootstrap, 2,000 resamples, percentile 2.5/97.5; (4) externally, calibration metrics are descriptive only — any deployment interpretation requires local recalibration, stated rather than performed.

From Material and Methods / Introduction (dataset roster), composed verbatim. TN-SCUI2020 provides class labels with delineations (dataset record in the main-text reference list); TN-SCUI2020 remains an optional extension pending authorised access.

From Results (Phase 1), verbatim. TTA effects are small and sign-mixed — the largest shift is +1.11 Dice points on TNUI-all, and DDTI moves slightly in the opposite direction — and DDTI is visibly the harder masked cohort (sensitivity 0.83, median HD95 roughly doubled), consistent with its lower zero-shot prototype scores. Best validation episodic Dice 0.7776 (epoch 13; early stop 21; 37.7 min). DDTI is the hardest masked cohort for both paradigms (supervised

0.7396; zero-shot 0.6782), consistent with its doubled boundary error (median HD95 ≈22 vs 10–11 px) and burned-in annotations.

From Results (Phase 3b, error propagation), verbatim. Error propagation is diffuse rather than confidence-localised: the oracle-minus-predicted AUROC gap is modest in the lowest-confidence quartiles, vanishes in Q3, and is descriptively largest in the *highest*-confidence quartile (+0.0589), the mean gap (≈0.022) closely matching the global T3 contrast. Segmentation error therefore imposes a diffuse ≈2-point tax on grounded features, and there is no confidence-gated "safe subgroup" in which predicted masks are output-equivalent to oracle boxes.

From Results (Phase 3, label-budget sweep), verbatim. The class-prototype label-budget sweep (K per class $\in \{5, 10, 20, 50, 200, \text{all}\}$, 25 draws) carries the classification-side annotation-efficiency claim: grounding dominates the identical whole-image budget at every K (+0.046 to +0.096 AUROC); R1p saturates by K=50 (0.6383 versus 0.6379 full-budget) and stays within 0.013–0.029 of the oracle, while R0 has not saturated even at K=200; and the ring is null-or-negative throughout (R2p ≤ R1p, R2g ≤ R1g at every K), corroborating the T4 null under a different head family. Absolute levels sit far below the linear heads (0.59–0.66 vs 0.85–0.91), as expected for a cosine prototype rule on frozen descriptors; the linear head remains the pre-declared primary classifier.

From Results (Phase 4, external family and abstention), verbatim. The primary external test is nominal only: on the single locked TNUI test+val pool JOINT-g exceeds ERM's sensitivity at the frozen operating point by 6.6 points (H1: +0.0661, 95% CI 0.005..0.128, p=0.033, beyond the family bar; McNemar 86 vs 60). The rank test is a flat null (H3: −0.011, p=0.858), and H2 is significant: joint grounding transfers its operating point *worse* than the frozen R1p by 5.3 points of pooled sensitivity (H2: −0.0534, CI −0.097..−0.010, p=0.011). The joint segmenter travels adequately (external Dice 0.674–0.765, empty-mask ≤ 0.3%), so the deficit lives in classification head-room, not mask collapse. JOINT-g had not saturated at the pre-registered 15-epoch cap (validation AUROC was still improving; the cap was not extended), so family-G magnitudes are conservative. Retaining the most-confident 70% of TN5000-test cases lifts JOINT-g's sensitivity at spec90 from 0.564 to 0.840, and at 50% coverage the curve reaches 0.874 in-domain and 0.897 on the external TNUI pool. The effect is referral-shaped, not accuracy-shaped: approximately 86–87% of the frozen point's malignant misses sit inside the abstained lower-confidence half on both panels (261 of 302 misses in-domain; 132 of 152 on pooled TNUI). No threshold or temperature was re-estimated for these numbers (raw-raw; the validation-fitted T=0.523 supplies only the confidence margin).

From Results (Phase 5, audit and cohort assembly), verbatim. Label coverage was complete for Batch 1 (6,005/6,005) and near-complete for Batch 2 (2,495/2,503), giving an overall parse rate of 0.9991 (8,500/8,508); the surviving join matched the corpus's own image map exactly (5,893/5,893 Batch-1 stems). Assembly gave 842 cases (601 + 241), and conservative whole-case near-duplicate screening excluded three, leaving 839 cases and 8,465 images (598 + 241), of which 522 are malignant (prevalence 0.622, inside the $[0.20, 0.85]$ gate). Frozen-row integrity held exactly (ERM re-hit 0.9401; the retrained R1p head reproduced 0.8889 against the locked 0.8902, within the 0.002 integrity tolerance; both JOINT arms matched to $\leq 10^{-6}$, temperatures 0.523/0.200), and segmentation QC was healthy (empty masks on 10.9% of images, 926/8,465; JOINT-g decoder 3.5% (293/8,465)). Mean predicted nodule areas were 4.1% and 5.9% of the frame respectively.

From Results (Phase 5, collapse and calibration detail), verbatim. Only ERM's interval clears 0.5 — by eight thousandths. AUPRC spans 0.608–0.666 against the 0.622 prevalence,

and shift drops from source are 0.31–0.39 AUROC for every linear head and 0.39 for ERM (drop 0.3921, the largest); even JOINT-img, modest in-domain (0.7207), loses 0.24, and the first-list-image view shows the same pattern (0.481–0.533). Calibration does not travel either: source-fitted temperatures *worsen* external ECE for five of six models (post-calibration ECE 0.26–0.31; only ERM improves, 0.2813 → 0.2145), and Brier scores (0.29–0.35) sit far above the source regime. In integers: of the 83 malignant cases R1p catches at the frozen point, ERM misses 77; of the 7 ERM catches, R1p misses 1 — an eleven-fold relative advantage, and the external case-level replication of the internal operating-point result.

From Results (Phase 6, integrity detail), verbatim. The seeded ThyUS-600 first-image panel (n=600, 376 malignant) was drawn from the licensed 839-case cohort and triangulated: 522 malignant, 8,465 images, prevalence 0.6222. Live probabilities reproduce the Phase-3/4 archives (ERM 0.9404 vs 0.9401; R1p 0.8889 vs 0.8889; JOINT arms to $\leq 2.25\times10^{-5}$ max $|\Delta p|$ against the 10^{-4} gate), and the 32-image ELA exactness probe confirmed the closed-form logit decomposition (max deviation R0 1.6×10^{-6}, R1p 2.9×10^{-7}, R2p 2.7×10^{-7}, JOINT-g 7.8×10^{-8}, JOINT-img 2.6×10^{-8}; gate 10^{-3}).

From Results (Phase 6, descriptives and faithfulness detail), verbatim. R1p-ELA attains a localisation ratio of 41.6 (median 27.8) where Grad-CAM attains 3.6. The decomposition marks R1p's precision as segmenter-grade (79.6% of attribution mass inside the box, normalised entropy 0.717) against ERM-family diffusion (in-box mass 0.146, entropy 0.928). The pointing game cuts the other way descriptively (Eigen-CAM 0.736 > R1p 0.637 > Score-CAM 0.580 > Grad-CAM 0.514): where a diffuse map happens to peak inside the box it scores like a concentrated one; the ratio and in-box-mass statistics instead report where the *bulk* of the explanation lies. Under shift, ERM's attention deserts the nodule: Grad-CAM's energy inside the frozen predicted mask falls from 0.1253 in-domain (≈3.2× the mask's area share) to 0.0549 on ThyUS-600 (≈1.3×). The reversal is carried by the random-order control, not by the guided curves: explanation-guided deletion degrades both models essentially identically (areas 0.3334 vs 0.3315), while random deletion erodes the diffuse image-wide classifier more slowly (random-control areas 0.4986 vs 0.5564). Insertion (descriptive, n=300) favours ERM (0.2043 vs 0.0351) with the same proviso, compounded by the out-of-distribution nature of sharp-lesion-on-blur hybrids for a mask-conditioned head.

From Results (Phase 6, joint ladder, stability, cascade diagnostics), verbatim. The deletion-gap ladder rises monotonically with the degree of built-in grounding (JOINT-img 0.041 < JOINT-g 0.102 < R1p 0.165). Under intensity noise R1p-ELA is the more stable explanation in both domains (mean Spearman 0.921 vs 0.832 on TN5000; 0.799 vs 0.755 on ThyUS-600), while affine warps invert the ordering (0.771 vs 0.871; 0.640 vs 0.810) — the two rows are the same concentration geometry read in opposite directions. ERM decorrelates below the bar only at its deepest stage (0.1205), and at the matching shallow stage (fc only, 0.2948) it fails identically: a 0.2 threshold at readout depth is unattainable for either model, rather than evidence of explanation defect. Independent resampled heads correlate at only 0.0920 on the same images, so the retained 0.2991 sits +0.2072 above that floor: the pixel map is *not* head-agnostic geometry. Meanwhile, the per-feature logit attribution decorrelates completely under head resampling (Spearman −0.0150, $|-|$ 0.0191; Pearson −0.0016).

From Discussion (mechanism and referral), verbatim. The parameter-free variant collapses to 0.47–0.57 Dice under zero-shot transfer, so the contribution sits in the decoder, not in nearest-prototype matching of the PANet kind. A small referral stream (≈30% of cases) removes the majority of the frozen point's false negatives without touching a single parameter. The descriptives cohere: 79.6% of the grounded map's mass lies inside the expert box versus 14.6%

for Grad-CAM. Committed diagnostics show the retention is neither head-agnostic geometry (random-vs-random floor 0.092; retained elevation +0.2072 above it) nor a broken explanation (per-feature attribution decorrelates completely).

From Results (Phase 4, grounding signal), verbatim. Under identical multi-task conditions the grounded head beats the whole-image head by 14.9 points (G3: +0.1489, CI 0.1136..0.1848).